

Network Fundamentals & OSI Model Report

Comprehensive Analysis of Wireshark Traffic Capture, OSI Layer Mapping, Visual Model, and WhatsApp Message Flow

Task 1: Wireshark Traffic Capture & Protocol Identification

Wireshark was installed on the primary network interface, and a live packet capture was initiated while browsing **instagram.com** via a standard web browser. After capturing traffic for approximately 30 seconds, the capture was stopped for analysis.

The following prominent protocol names were observed in the Wireshark packet list:

- **DNS (Domain Name System):** Resolved `instagram.com` domain names to corresponding server IP addresses.
- **TLSv1.3 (Transport Layer Security):** Handled secure, encrypted cryptographic handshake sessions for secure HTTPS browsing.
- **HTTP/2:** Managed high-performance multiplexed application data transfer for web assets (images, scripts, styles).
- **TCP (Transmission Control Protocol):** Provided reliable, connection-oriented socket streams (SYN, ACK, FIN flags).
- **QUIC:** Utilized UDP-based transport for fast connection setup and encrypted media streaming.
- **IPv4 / IPv6:** Handled logical network addressing and packet routing across intermediate routers.
- **ARP (Address Resolution Protocol):** Resolved local IP addresses to MAC addresses on the local area network (LAN).
- **DHCP / ICMP:** Managed dynamic IP lease configuration and network diagnostics (ping/echo requests).

Task 2: Packet Analysis & OSI Layer Mapping

Three representative packets from the Wireshark capture session were selected and mapped to their corresponding OSI reference model layers:

Packet No.	Protocol	Wireshark Info Details	OSI Layer Representation
Packet #12	DNS	Standard query A instagram.com	Application Layer (Layer 7) Initiates domain name resolution query for human-readable web addresses.
Packet #15	TCP	[SYN] Seq=0 Win=64240 Len=0 MSS=1460	Transport Layer (Layer 4) Manages reliable connection establishment and sequencing between client and server.

Packet No.	Protocol	Wireshark Info Details	OSI Layer Representation
Packet #18	IPv4	Destination: 157.240.229.174 (Instagram Server)	Network Layer (Layer 3) Handles logical addressing and routing of packets across interconnected internet nodes.

Task 3: Colorful OSI Model & Real-World App Examples

The Open Systems Interconnection (OSI) model standardizes network communication functions across 7 distinct conceptual layers, illustrated below with real-world applications:

Layer	OSI Layer Name	Core Function	Real-World Application Example
Layer 7	Application Layer	End-user software services & network access.	WhatsApp / Gmail / Chrome: Chat UI, composing email, rendering web pages.
Layer 6	Presentation Layer	Data translation, encryption, and compression.	TLS / SSL / JPEG / PNG: Encrypting chat payloads and compressing image formats.
Layer 5	Session Layer	Dialog control, session establishment, and teardown.	HTTPS / NetBIOS: Maintaining active login sessions or video conference channels.
Layer 4	Transport Layer	Reliable end-to-end data delivery and flow control.	TCP / UDP: Ensuring all WhatsApp message packets arrive intact in correct order.
Layer 3	Network Layer	Logical addressing and packet routing across networks.	IPv4 / IPv6 / Routers: Directing data packets from sender to Instagram servers globally.
Layer 2	Data Link Layer	Node-to-node physical addressing and error detection.	Wi-Fi (802.11) / Ethernet / MAC: Framing data across local network switches and access points.
Layer 1	Physical Layer	Transmission of raw binary bits over physical media.	Fiber Optics / Copper Cables / Radio Waves: 5G cellular signals and Wi-Fi radio frequencies.

Task 4: Journey of a WhatsApp Message Through the OSI Model

When a user types and sends a message (e.g., "Hello!") on WhatsApp, the data travels downward through the OSI layers before transmission, and upward at the destination. Below is the step-by-step description of what happens at 4 key layers:

1. Application Layer (Layer 7)

The user interacts with the WhatsApp chat interface, typing the text and pressing send. The application formats this user action into an application-layer data payload using messaging protocols (such as XMPP or WhatsApp's custom Protobuf binary protocol over WebSockets).

2. Transport Layer (Layer 4)

WhatsApp utilizes robust transport protocols (often TCP or customized QUIC/UDP). The Transport layer takes the message data stream, breaks it down into manageable segments or packets, adds sequence numbers, and establishes reliable end-to-end error checking. If any packet is lost in transit, the Transport layer ensures it is retransmitted, guaranteeing that *"Hello!"* arrives completely intact.

3. Network Layer (Layer 3)

The Network layer encapsulates the transport segments into packets (IP datagrams). It attaches the sender's IP address and the destination WhatsApp server's IP address. Routers across the internet examine these IP headers to determine the optimal path through global networks to forward the packet toward its destination.

4. Physical & Data Link Layers (Layers 1 & 2)

At the lowest level, the Data Link layer places the packet into local frames with MAC addresses, passing it to the Physical layer. Here, the binary data is converted into physical signals—such as Wi-Fi radio frequency waves emitted by your phone's antenna to your home router, or electrical pulses traveling across fiber optic cables to internet exchange points.